

Quantitative Aptitude – Complete Notes

Important formulas + easy examples for placement & competitive exams

1. Number System

Key concepts

Natural numbers: 1,2,3...; Whole numbers: 0,1,2...; Integers: ...-2,-1,0,1,2...; Prime number has exactly two factors.

Divisibility rules

2: last digit even. 3: sum of digits divisible by 3. 5: last digit 0/5. 9: digit sum divisible by 9. 11: difference of alternating digit sums is 0 or a multiple of 11.

HCF & LCM

For two positive integers: $HCF \times LCM = \text{Product of the numbers}$.

Example

Find HCF of 24 and 36. Factors: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, so $HCF = 2^2 \times 3 = 12$. $LCM = 2^3 \times 3^2 = 72$.

2. Percentage

Formula

$\text{Percentage} = (\text{Part/Whole}) \times 100$. $\text{Percentage increase} = \text{Increase/Original} \times 100$.

Useful result

If a value increases by x%, new value = $\text{Original} \times (100+x)/100$. If it decreases by x%, new value = $\text{Original} \times (100-x)/100$.

Example

A student scores 72 out of 90. $\text{Percentage} = 72/90 \times 100 = 80\%$.

3. Profit, Loss & Discount

Formula

$\text{Profit} = \text{SP} - \text{CP}$; $\text{Loss} = \text{CP} - \text{SP}$; $\text{Profit\%} = \text{Profit/CP} \times 100$; $\text{Loss\%} = \text{Loss/CP} \times 100$; $\text{Discount} = \text{MP} - \text{SP}$; $\text{Discount\%} = \text{Discount/MP} \times 100$.

Example

$\text{CP} = \text{₹}800$, $\text{SP} = \text{₹}920$. $\text{Profit} = \text{₹}120$; $\text{Profit\%} = 120/800 \times 100 = 15\%$.

Discount example

$\text{MP} = \text{₹}1,000$ and discount = 20%. $\text{SP} = 1000 \times 80/100 = \text{₹}800$.

4. Simple Interest (SI) & Compound Interest (CI)

Formula

$\text{SI} = \text{P} \times \text{R} \times \text{T}/100$; $\text{Amount} = \text{P} + \text{SI}$.

CI formula

For annual compounding: $A = P(1+R/100)^T$ and $CI = A - P$.

Example

$P = \text{₹}5,000$, $R = 10\%$, $T = 2$ years. $SI = 5000 \times 10 \times 2 / 100 = \text{₹}1,000$. $CI = 5000(1.1)^2 - 5000 = \text{₹}1,050$.

5. Ratio & Proportion

Concept

Ratio compares quantities: $a:b$. Proportion means two ratios are equal: $a:b = c:d$, so $ad = bc$.

Example

Divide $\text{₹}600$ in ratio $2:3$. Total parts = 5. Shares = $\text{₹}600 \times 2/5 = \text{₹}240$ and $\text{₹}600 \times 3/5 = \text{₹}360$.

6. Time & Work

Formula

If A completes work in x days, one-day work = $1/x$. Combined work = sum of individual one-day works.

Example

A does a job in 10 days and B in 15 days. Together: $1/10 + 1/15 = 1/6$, so they finish in 6 days.

7. Time, Speed & Distance

Formula

Speed = Distance/Time; Distance = Speed×Time; Time = Distance/Speed.

Conversion

$1 \text{ m/s} = 18/5 \text{ km/h}$; $1 \text{ km/h} = 5/18 \text{ m/s}$.

Example

A car travels 180 km in 3 hours. Speed = $180/3 = 60 \text{ km/h}$.

8. Probability

Formula

Probability = Favorable outcomes / Total equally likely outcomes. $0 \leq P(E) \leq 1$.

Example

A fair die is rolled. Probability of getting an even number = $3/6 = 1/2$.

9. Permutation & Combination

Formula

$nPr = n!/(n-r)!$; $nCr = n!/[r!(n-r)!]$.

Meaning

Permutation: arrangement/order matters. Combination: selection/order does not matter.

Example

Arrange 3 different books: $3! = 6$ ways. Choose 2 students from 5: ${}^5C_2 = 10$ ways.

10. Average

Formula

Average = Sum of observations / Number of observations.

Example

Average of 10, 20, 30, 40 = $100/4 = 25$.

Shortcut

If every value increases by k , average also increases by k .

11. Algebra & Linear Equations

Identities

$(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$; $a^2 - b^2 = (a-b)(a+b)$.

Example

Solve $3x+5=20$. $3x=15$, therefore $x=5$.

12. Mixture & Alligation

Concept

Alligation helps find the ratio in which two quantities/prices are mixed to obtain a desired average price.

Example

Rice ■40/kg and ■60/kg are mixed to get ■50/kg. Since 50 is midway, ratio = 1:1.

13. Ages

Method

Represent present ages with variables and form equations from the given relationships.

Example

A is 5 years older than B. If $B=12$, $A=17$ years.

14. Partnership

Formula

Profit share is proportional to Capital $\times$ Time.

Example

A invests ■10,000 for 12 months; B ■15,000 for 8 months. Both have 120,000 capital-months, so profit ratio = 1:1.

15. Boats & Streams

Formula

Downstream speed = Boat speed + Stream speed. Upstream speed = Boat speed – Stream speed.

Example

Boat speed 10 km/h, stream 2 km/h: downstream = 12 km/h, upstream = 8 km/h.

16. Data Interpretation (DI)

Types

Table DI, bar graph, line graph, pie chart and caselet DI.

Approach

Read units carefully → identify required data → calculate totals/ratios/percentages → estimate when options permit.

Example

Sales are 100, 150 and 250 units. Total = 500. Second month's share = $150/500 \times 100 = 30\%$.

17. Mensuration

Formula

Rectangle: Area= $l \times b$, Perimeter= $2(l+b)$. Square: Area= a^2 , Perimeter= $4a$. Circle: Area= πr^2 , Circumference= $2\pi r$.

Example

Rectangle 12 m × 5 m: area = 60 m²; perimeter = 34 m.

18. Geometry

Key facts

Triangle angle sum = 180°. Quadrilateral angle sum = 360°. Pythagoras in a right triangle: hypotenuse² = base² + perpendicular².

Example

Right triangle with sides 3 and 4: hypotenuse = $\sqrt{9+16}=5$.

19. Simplification & Approximation

BODMAS

Brackets → Orders → Division → Multiplication → Addition → Subtraction.

Example

$20 + 10 \times 2 = 20 + 20 = 40$, not 60.

20. Data Sufficiency – Basic Approach

Method

Check whether the given statements provide enough information to answer the question. Do not calculate more than necessary.

Tip

Test statement I alone, statement II alone, then both together.

Quick Formula Revision Sheet

Topic	Main formula
Percentage	$\text{Part/Whole} \times 100$
Profit%	$\text{Profit/CP} \times 100$
Loss%	$\text{Loss/CP} \times 100$
SI	$P \times R \times T / 100$
CI	$P(1+R/100)^T - P$
Ratio	a:b; proportion $a/b=c/d$
Work	One-day work = $1/\text{days}$
Speed	Distance/Time
Probability	Favourable/Total outcomes
Permutation	$nPr = n!/(n-r)!$
Combination	$nCr = n!/[r!(n-r)!]$
Average	Sum/Number
Rectangle area	$l \times b$
Circle area	πr^2

Note: These notes cover the major Quantitative Aptitude topics with basic formulas and examples. For exam preparation, practice topic-wise questions after learning each formula.